

Mapping Tree Canopy and Tree Equity in Calgary: Visualization and Analysis using Open Data

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Data For Good Calgary



About Me

- Started career in civil engineering and transportation project management, then pursued computer science to combine engineering, technology, and data.
- Work at Development Seed and serve as a Technical Lead supporting NASA's Office of Data Science and Informatics (ODSI) on the Commercial Satellite Data Acquisition program
- Passionate about using data and technology to better understand cities and help make them more accessible, vibrant, and exciting places to live.

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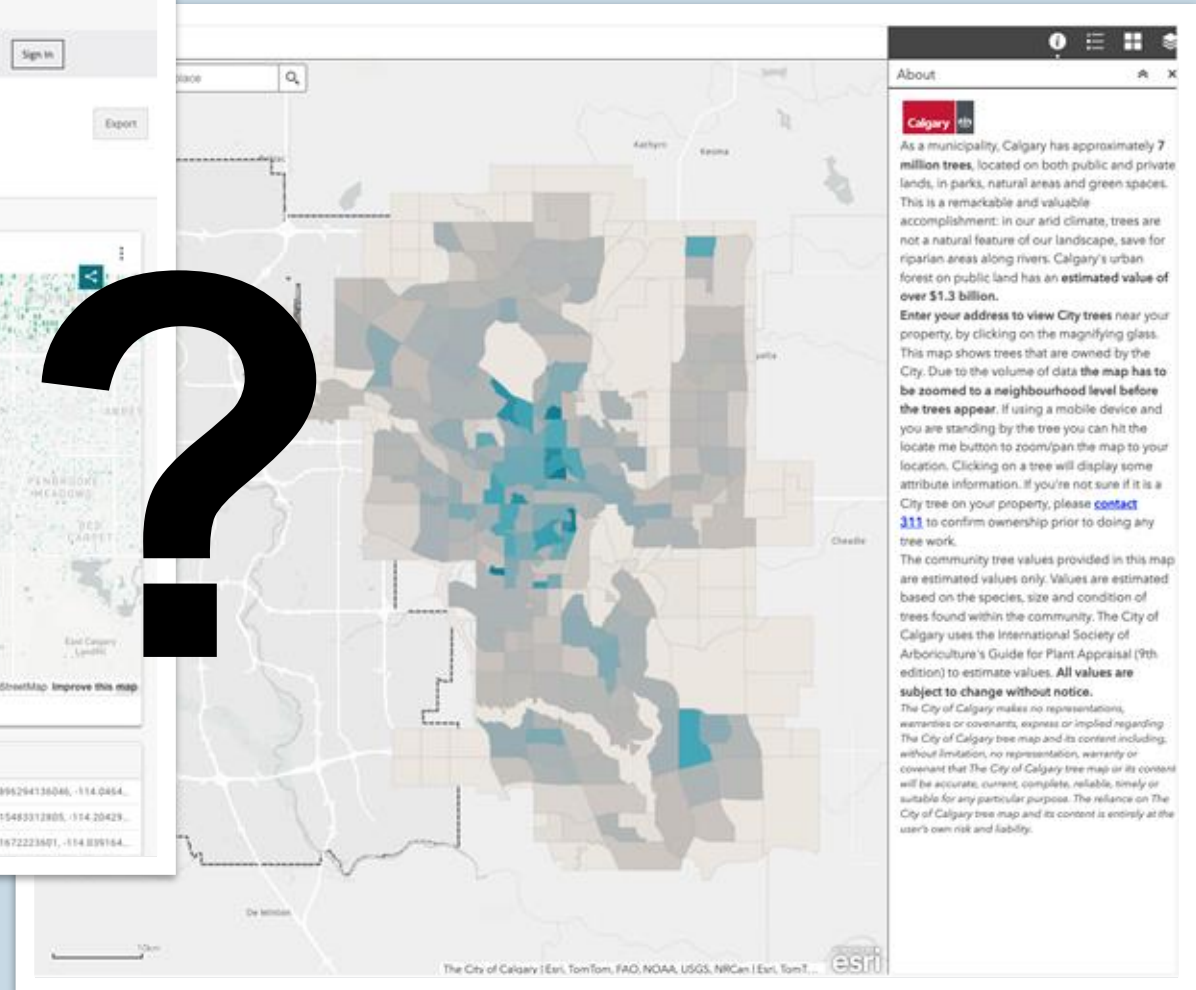
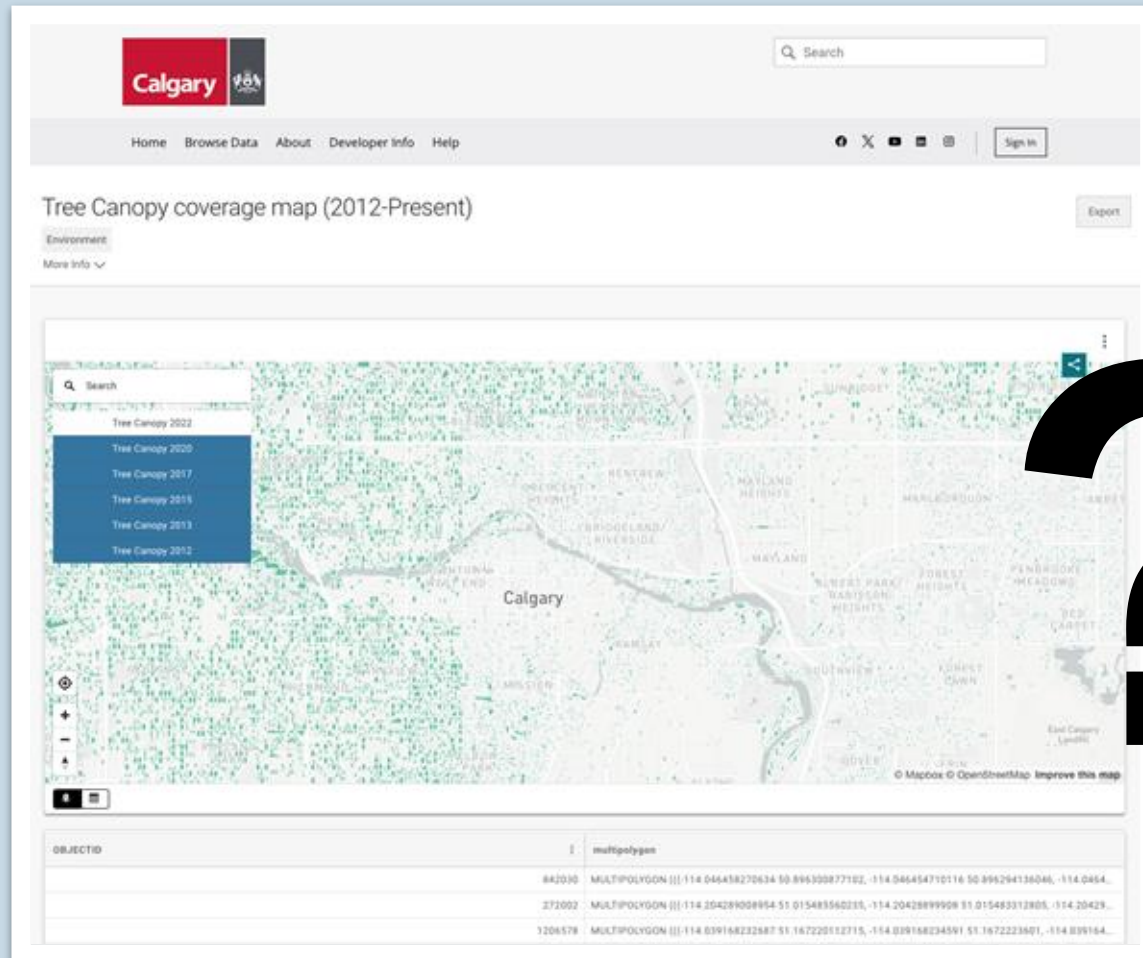


Calgary Climate Hub

- Volunteer-led, non-profit organization working towards meaningful local action on climate change
- Climate Hub programs:
 - Forests for Calgary (Tiny Forests, Miyawaki Method)
 - Advocacy and campaigns
 - Tree Equity



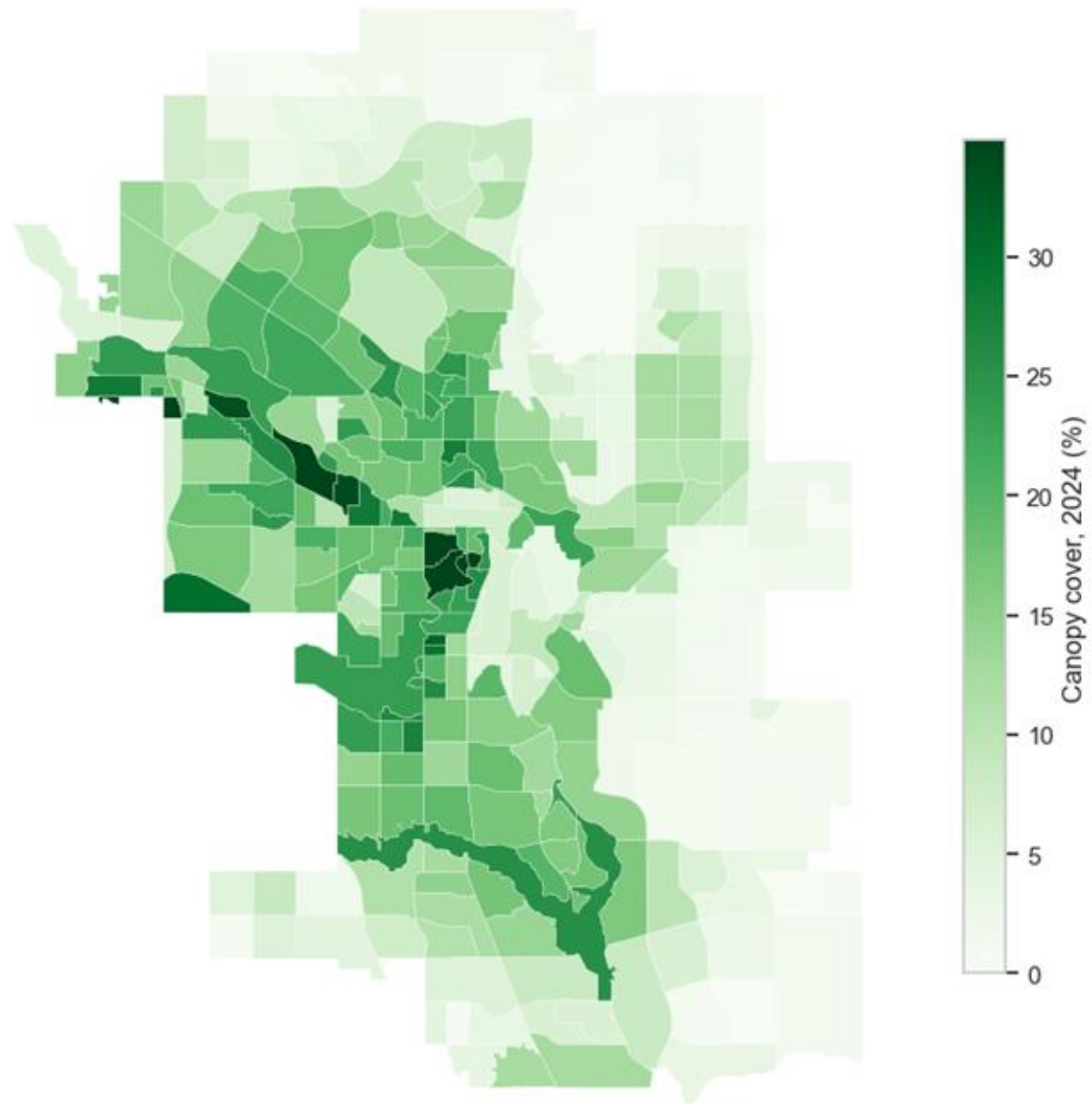
Where should we plant trees?



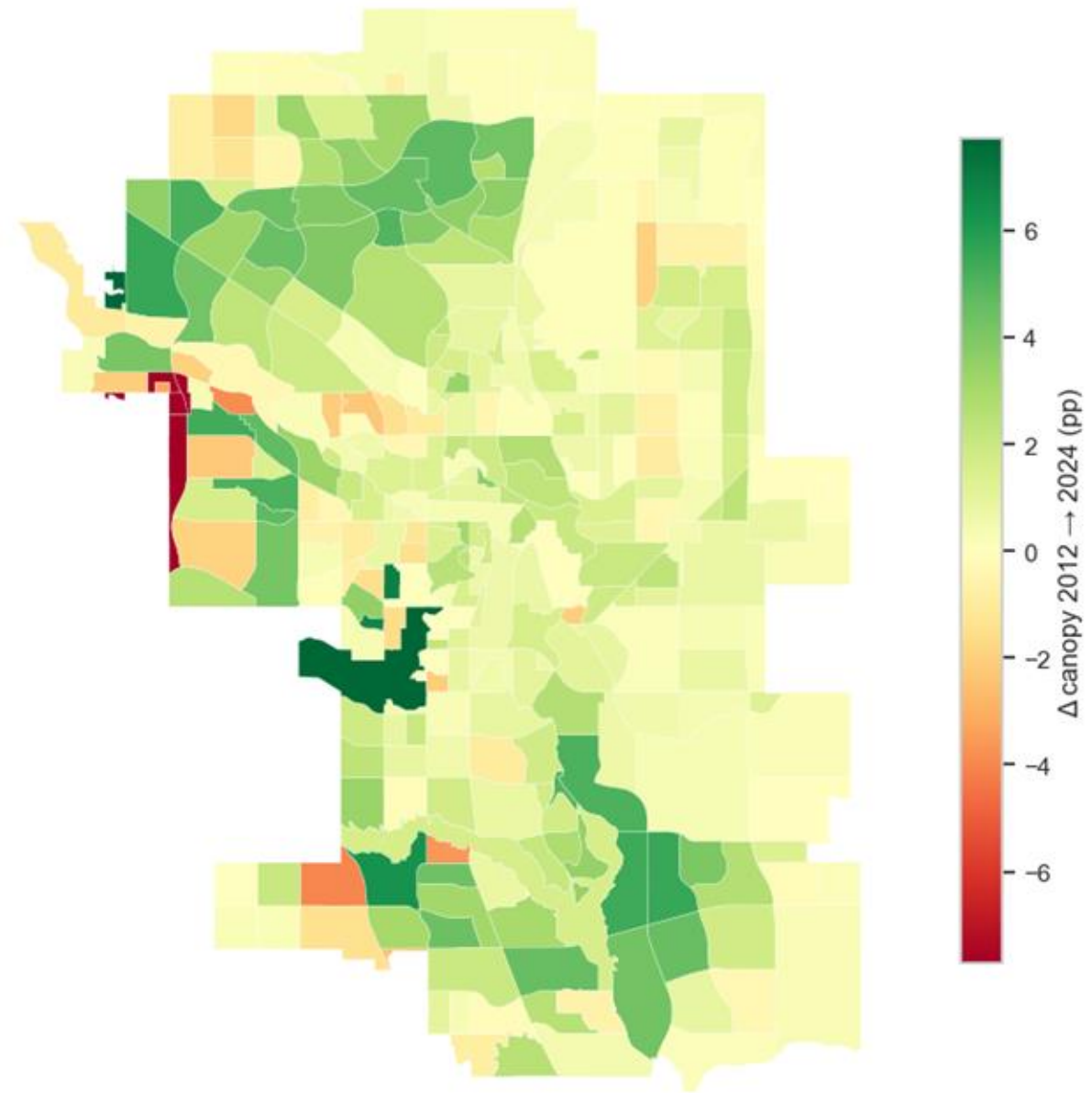
Calgary Tree Canopy

- 9.26% canopy (2024) -> 16% target
- Policy Documents:
 - **Calgary Parks Plan:** 16% by 2060
 - **Calgary Climate Strategy, Pathways to 2050:** 10% by 2030, 16% by 2050
 - **2023-2026 Climate Implementation Plan:** 14-20% by 2058
- Lowest tree canopy in Northeast Calgary (NE), highest concentration of visible minorities

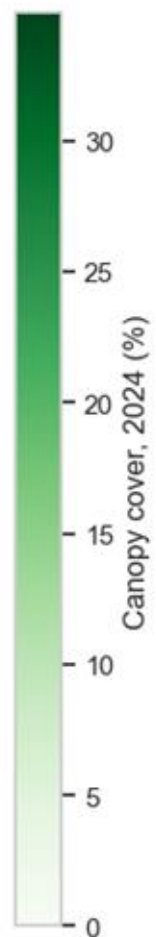
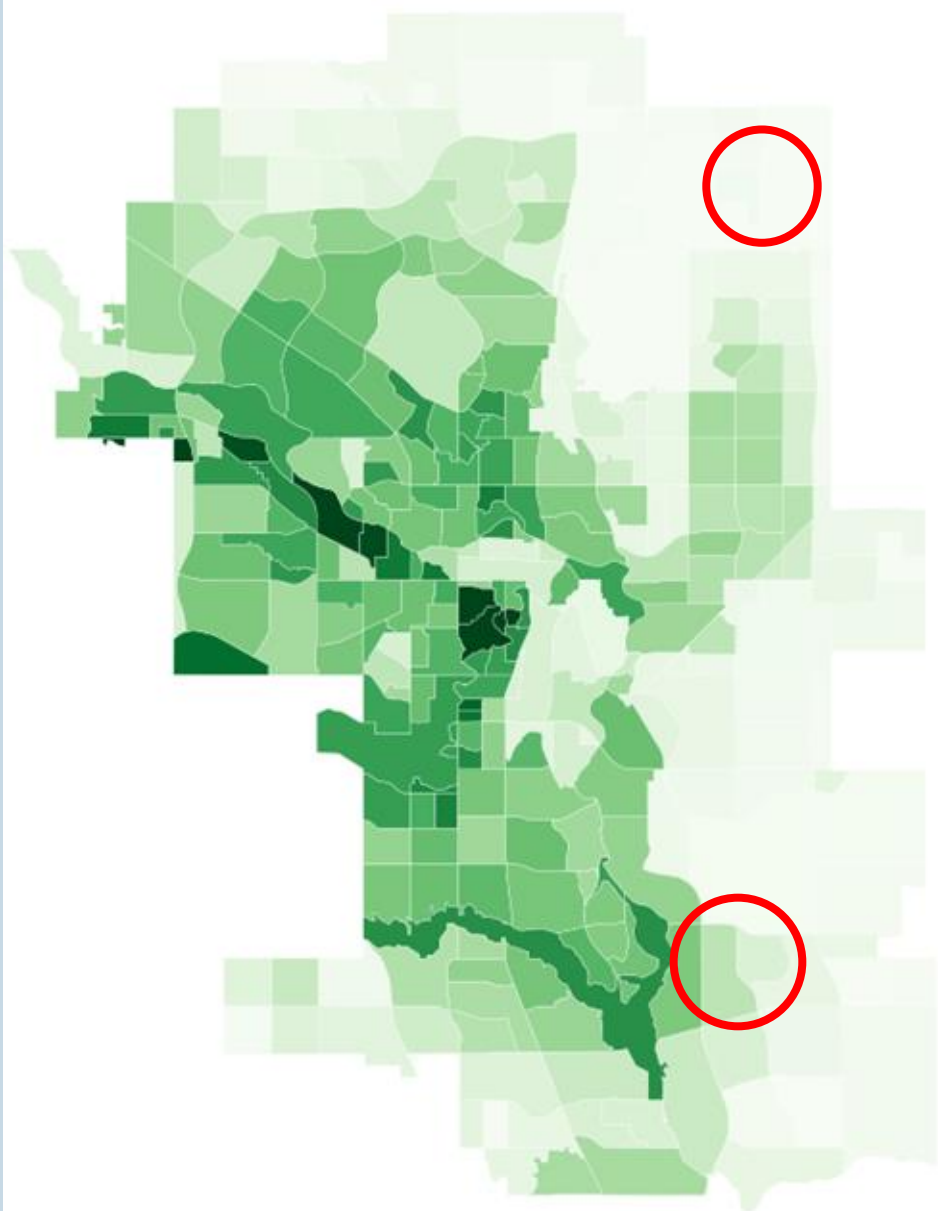
Canopy cover by community, 2024



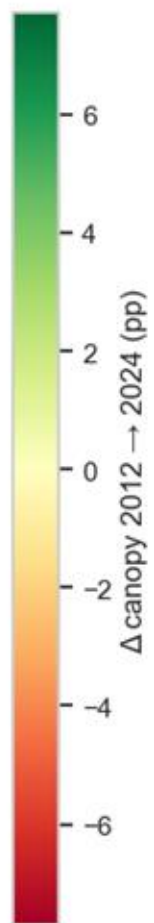
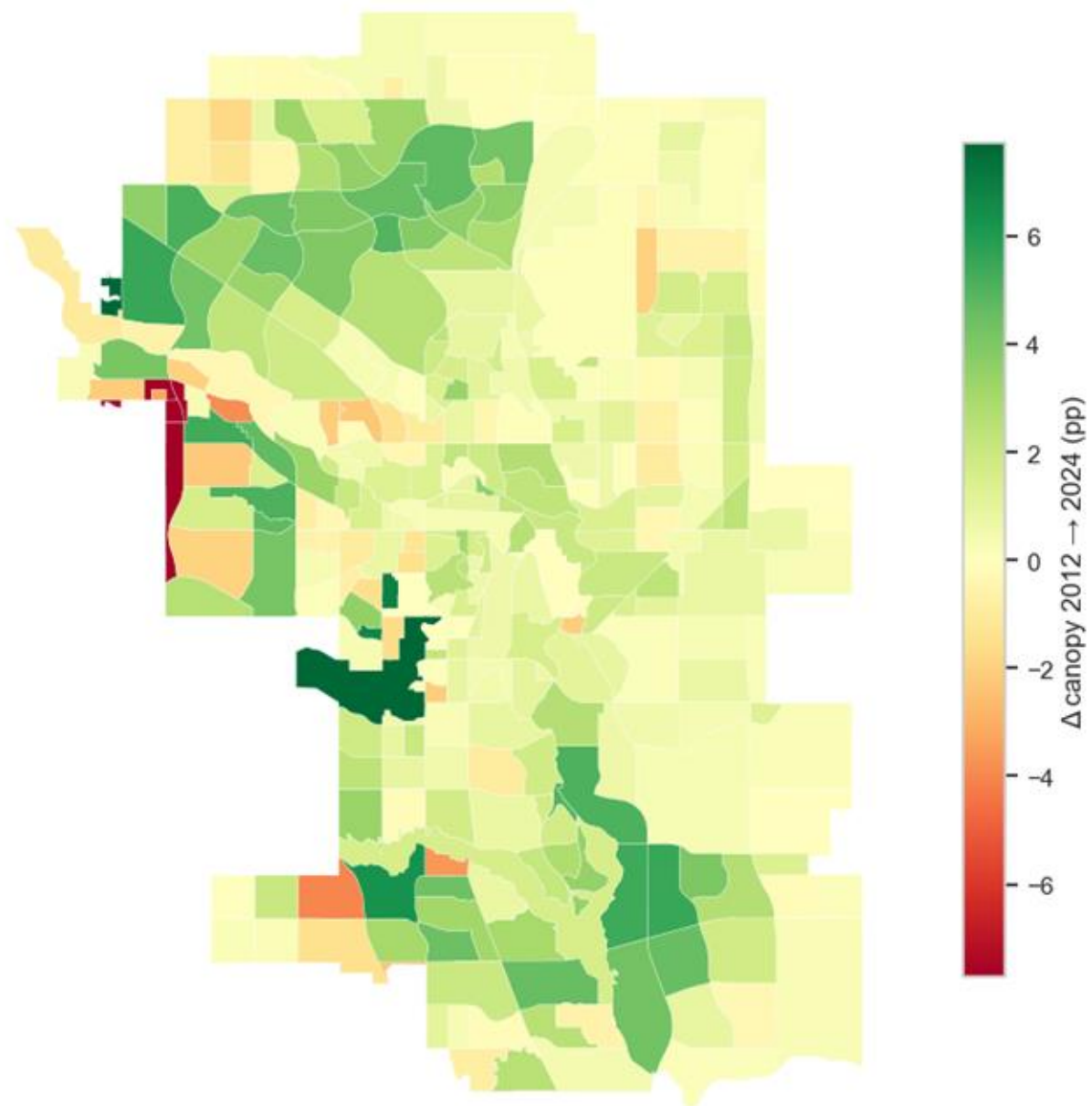
Canopy change by community, 2012 → 2024



Canopy cover by community, 2024



Canopy change by community, 2012 → 2024



McKenzie Towne



2001



2010

Saddle Ridge



2001

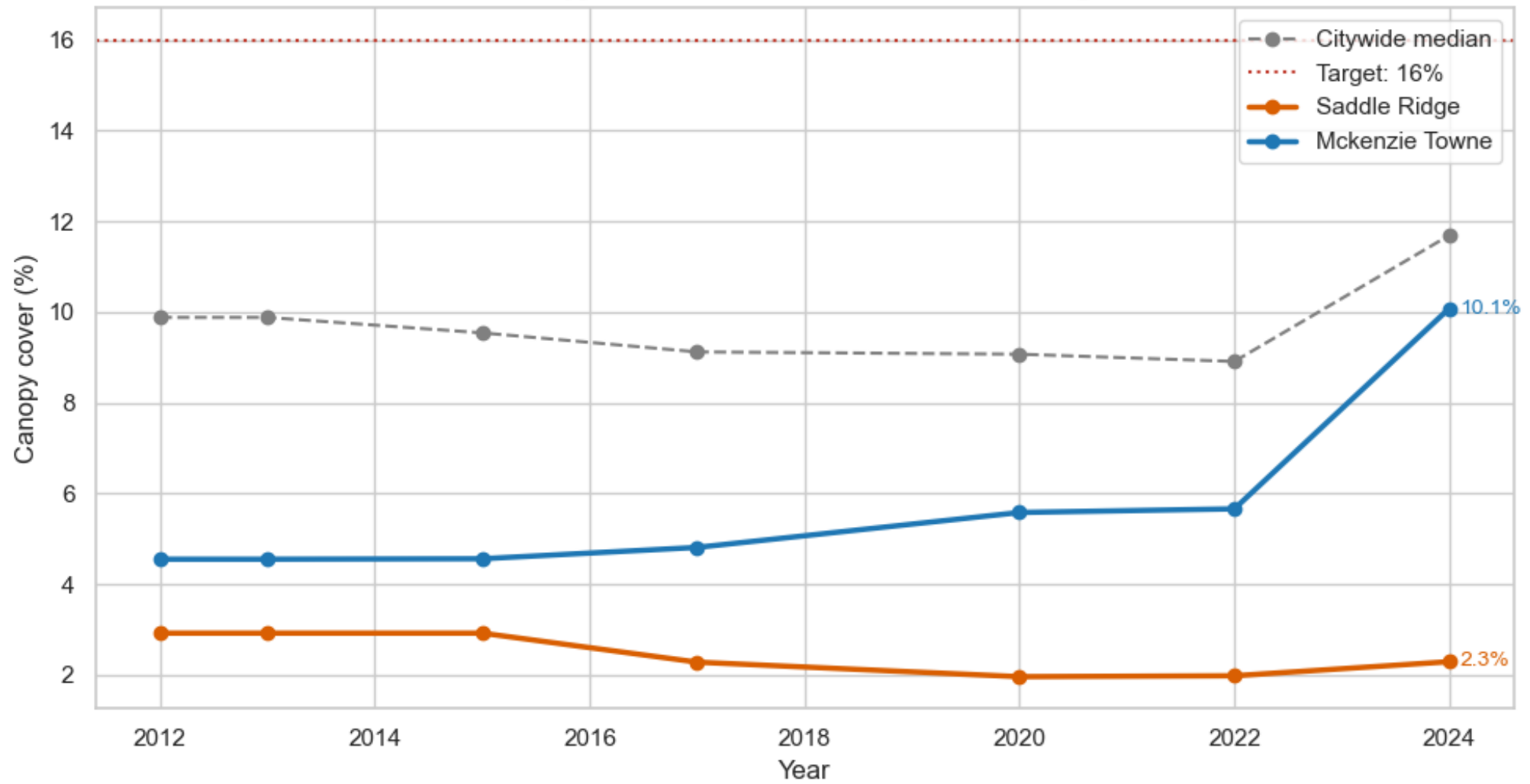


2010





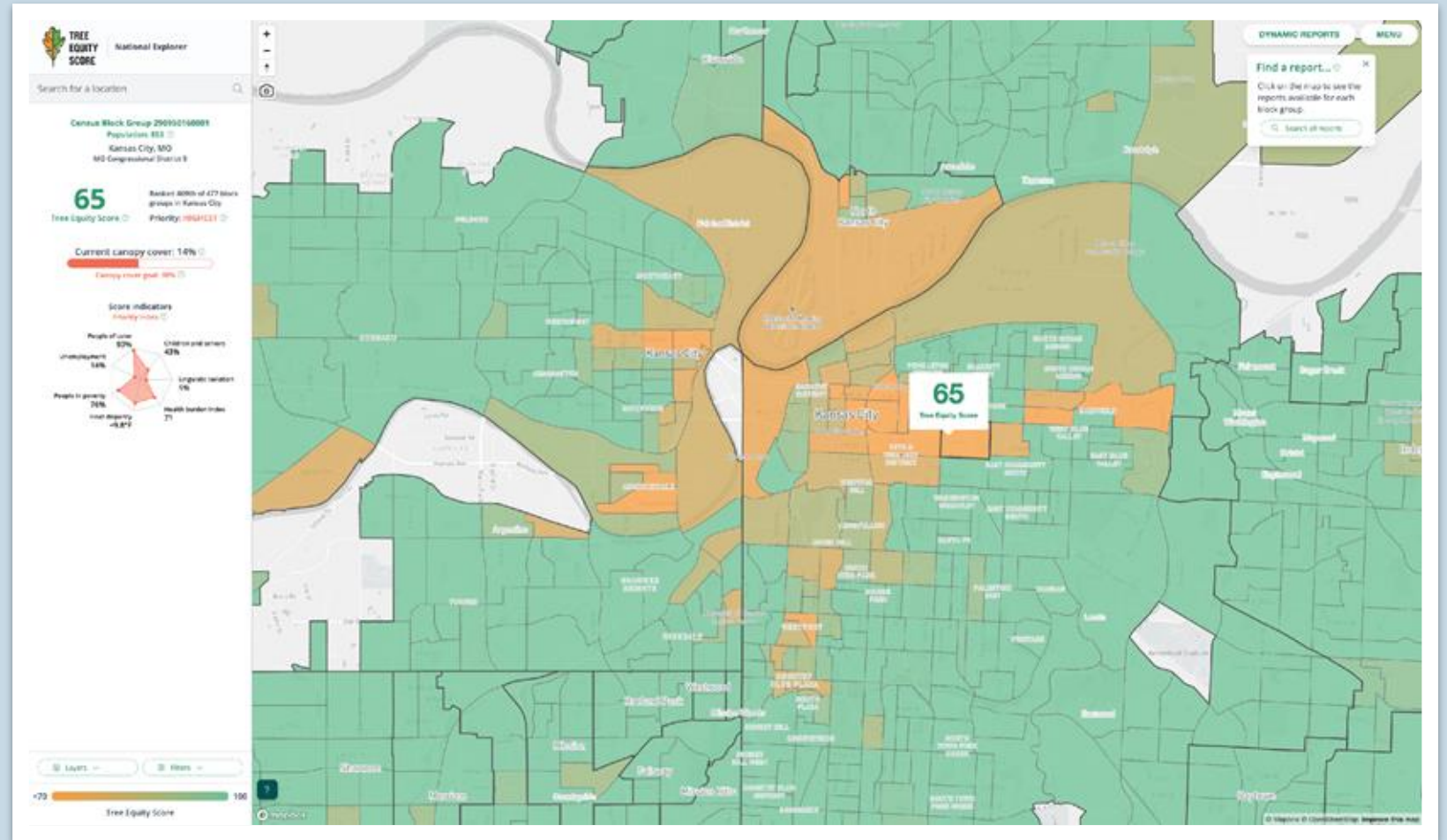
Canopy Cover — McKenzie Towne vs Saddle Ridge



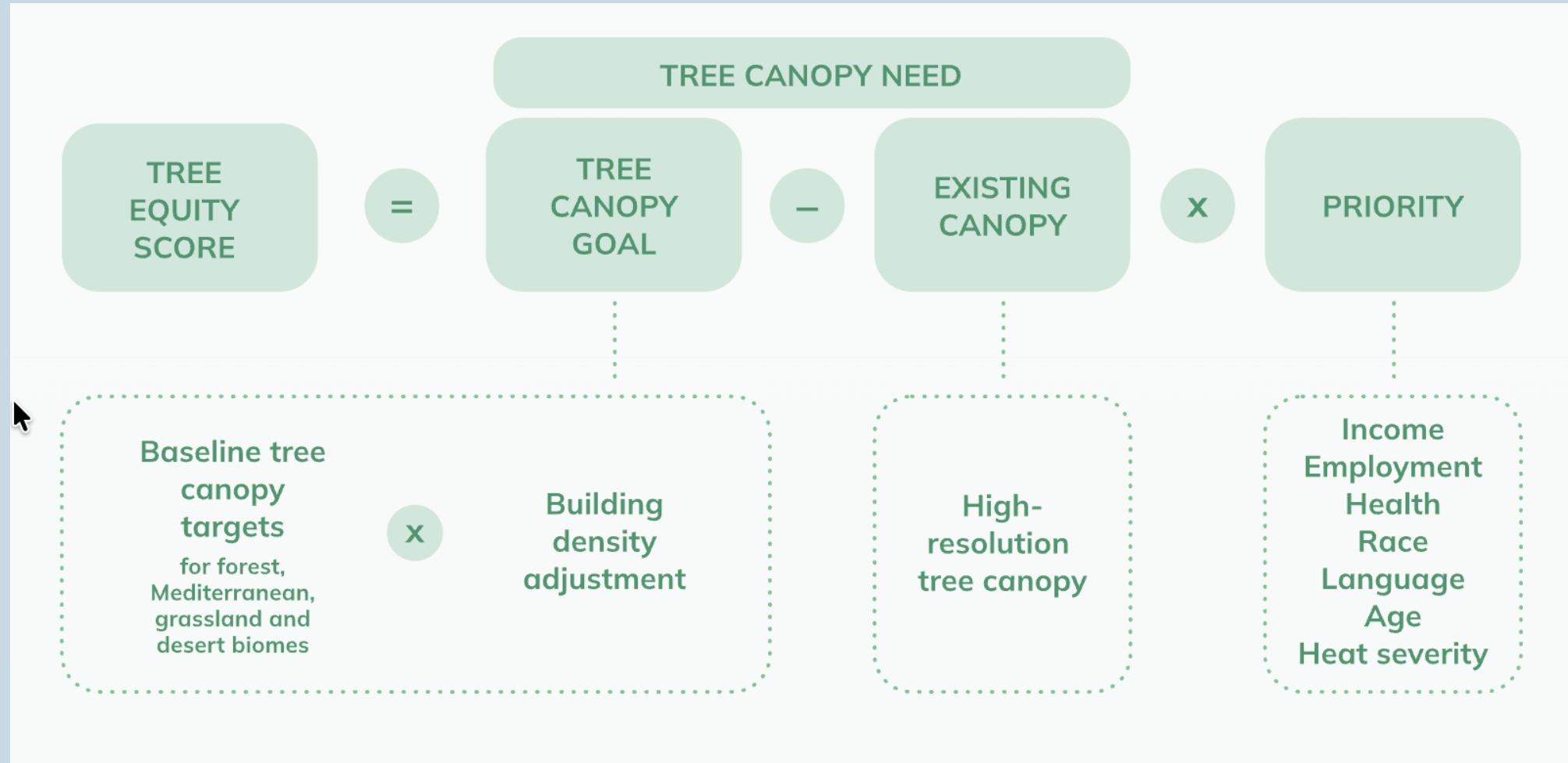


American Forests Tree Equity Score

- Measures whether everyone in a neighbourhood benefits from tree canopy
- High resolution maps of tree equity scores across USA
- Well established methodology, also adopted by City of Toronto

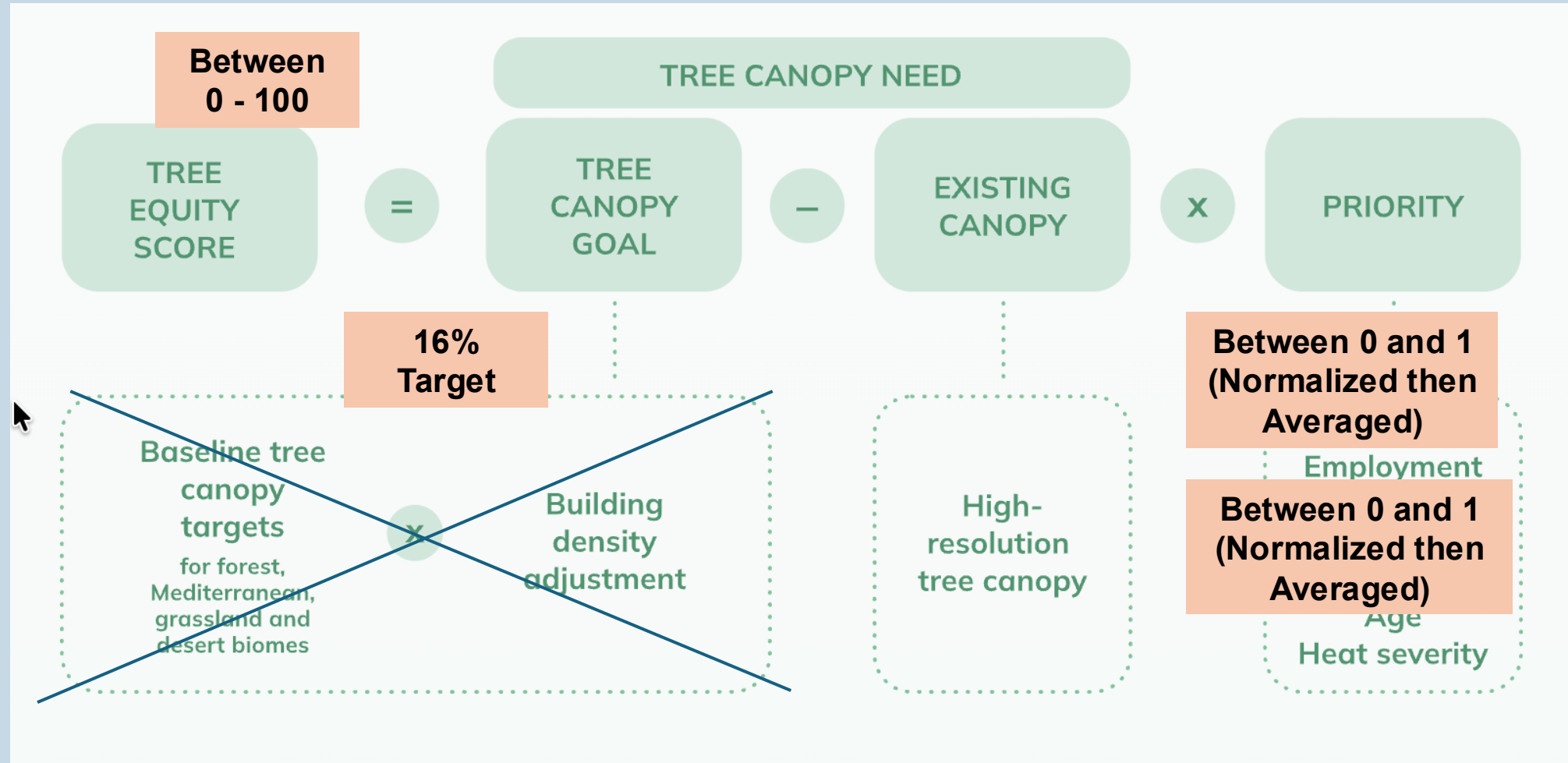


Methodology

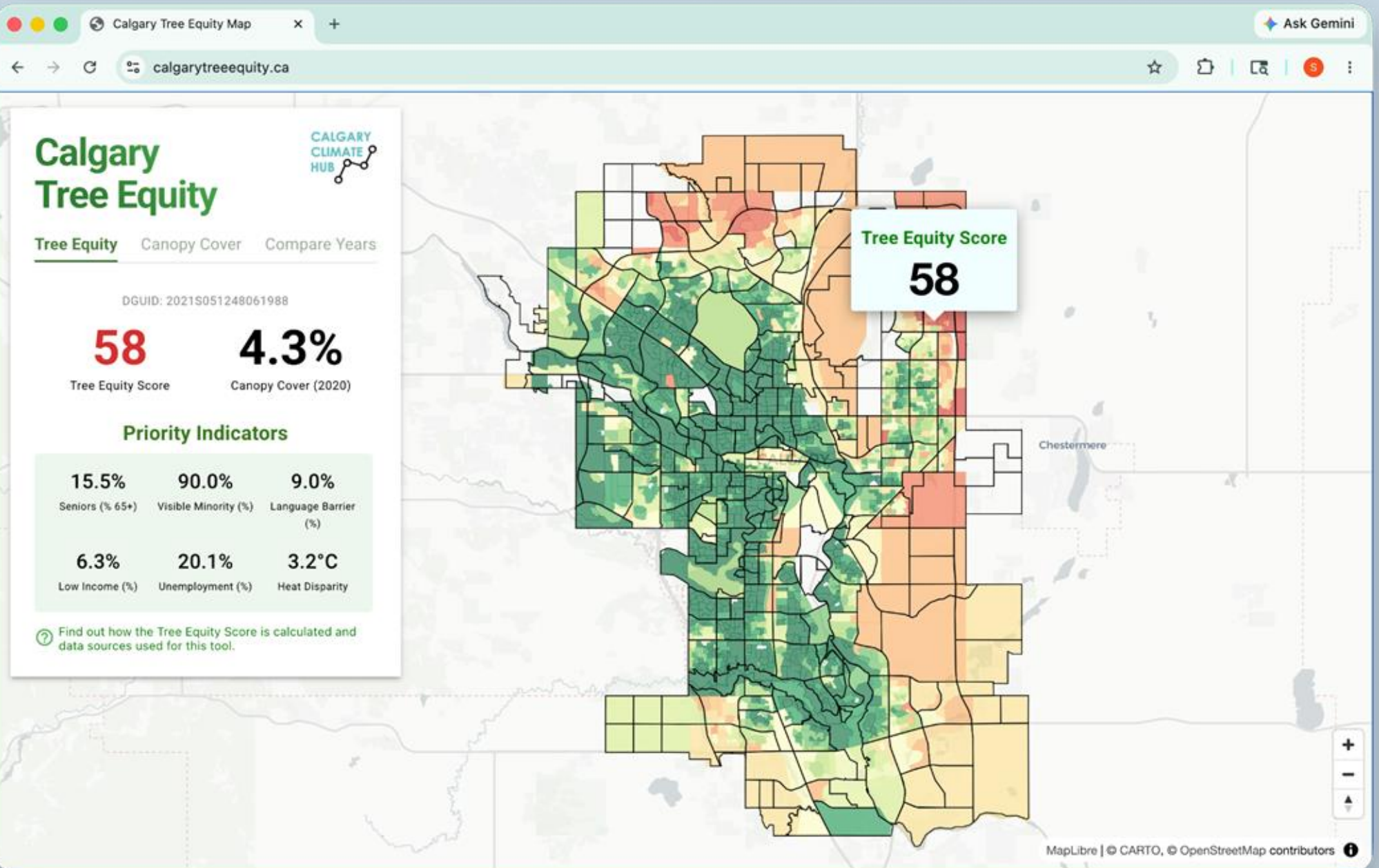


Source: <https://www.treeequityscore.org/methodology>

Methodology



Source: <https://www.treeequityscore.org/methodology>



Open Data and Open Source

- Data Sources
 - LiDAR tree canopy outlines
 - Aerial photography (orthophoto)
 - Statistics Canada 2021 Census Profile
 - Landsat Multi-spectral remote sensing data
- Open and reproducible analysis and visualization
 - No software or data licenses

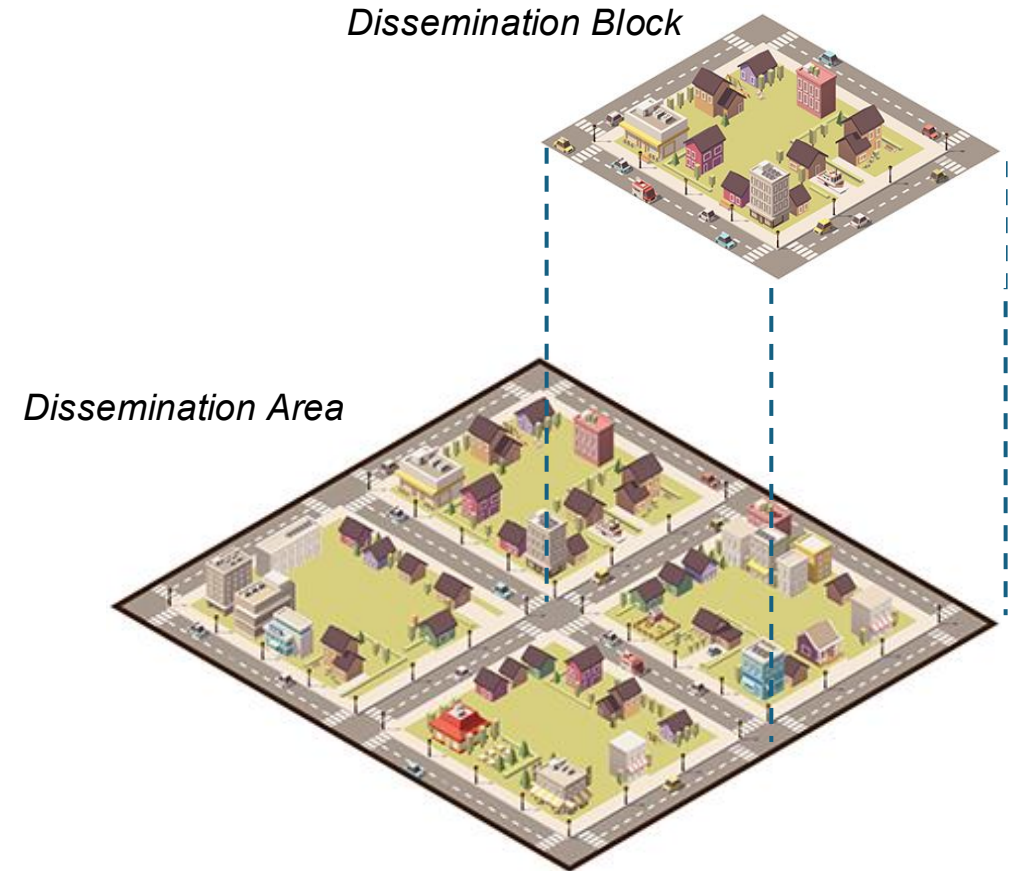


PMtiles



Statistics Data

- Priority Indicators
 - Seniors (65+) and Children (0-4)
 - Visible Minority
 - Language Isolation
 - Low Income (LIM-AT)
 - Unemployment Rate
- Geography
 - Census Dissemination Area (boundary of 400 to 700 residents)
- Access using The Statistical Data and Metadata eXchange (SDMX) standard



Land Surface Temperature

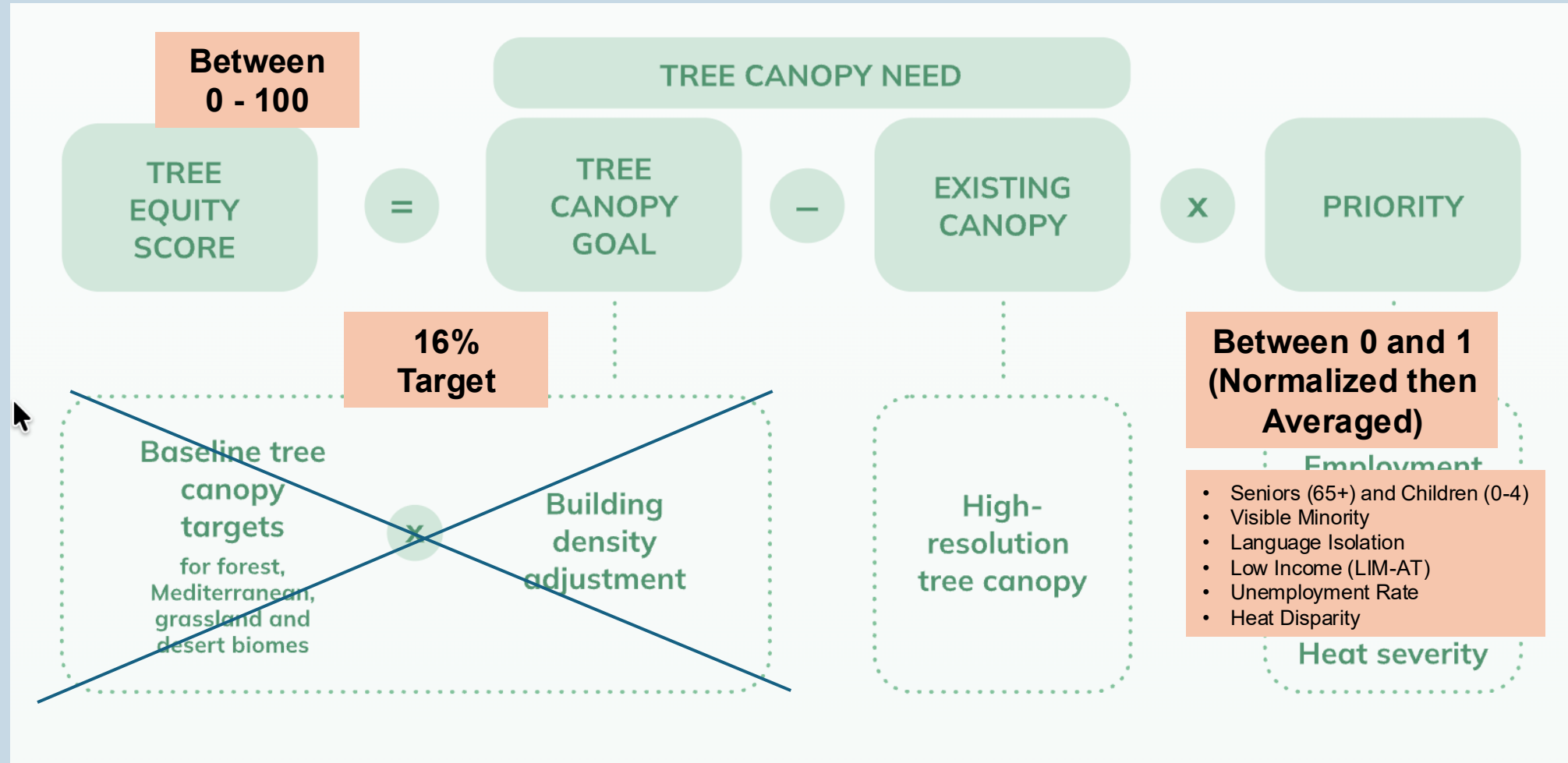
- Spatio Temporal Asset Catalog (STAC), open metadata standard for geospatial data
- Microsoft Planetary Computer, open catalog of remote sensing, environmental data

```
catalog = pystac_client.Client.open(
    "https://planetarycomputer.microsoft.com/api/stac/v1",
    modifier=planetary_computer.sign_inplace,
)

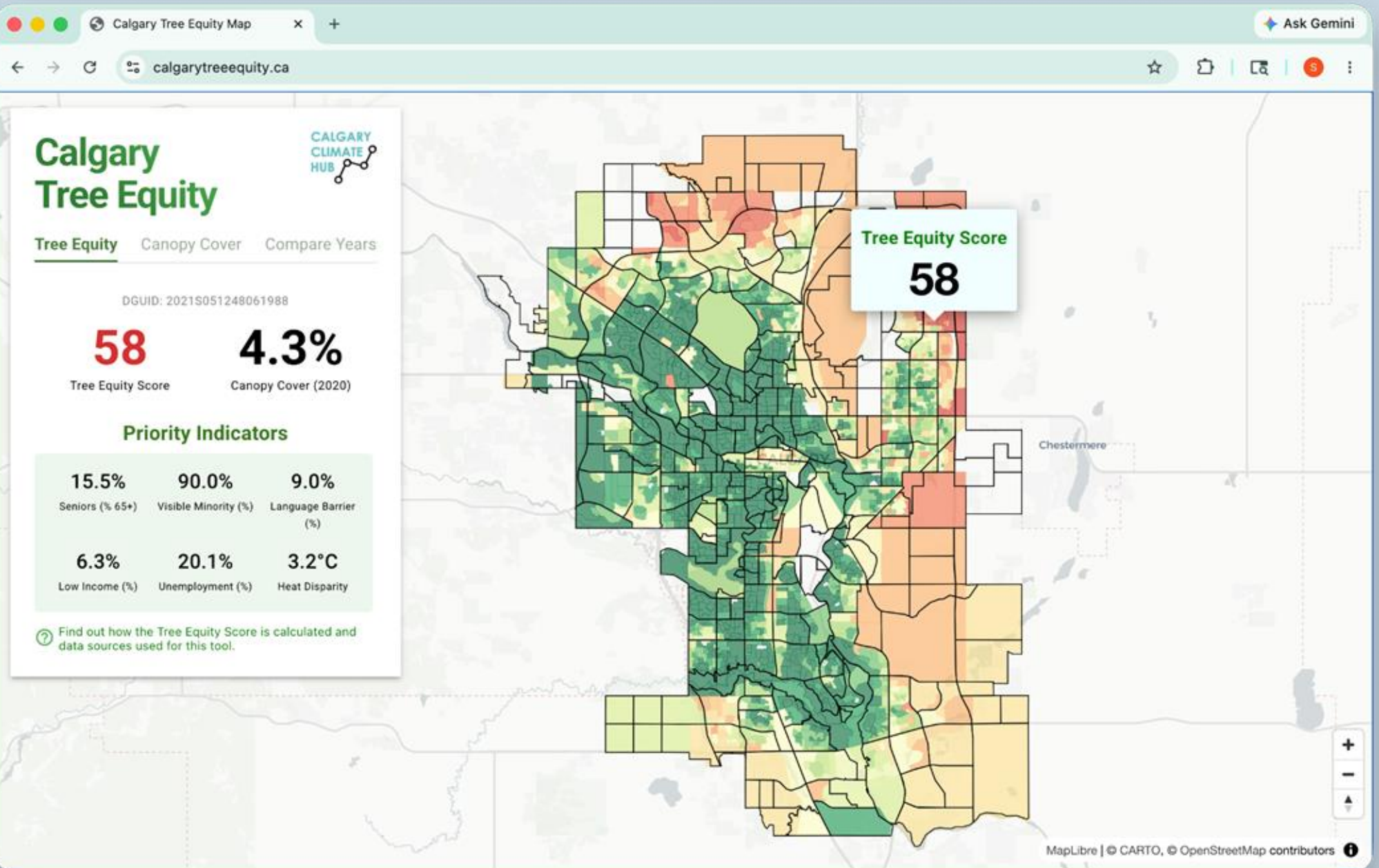
time_range = "2020-06-01/2020-07-30"

search = catalog.search(
    collections=["landsat-c2-l2"],
    bbox=boundary_bbox,
    datetime=time_range,
    query={"eo:cloud_cover": {"lt": 1}},
)
```

Methodology



Source: <https://www.treeequityscore.org/methodology>



Calgary Tree Equity



Tree Equity Canopy Cover Compare Years

Year

2020

Canopy Cover (%)

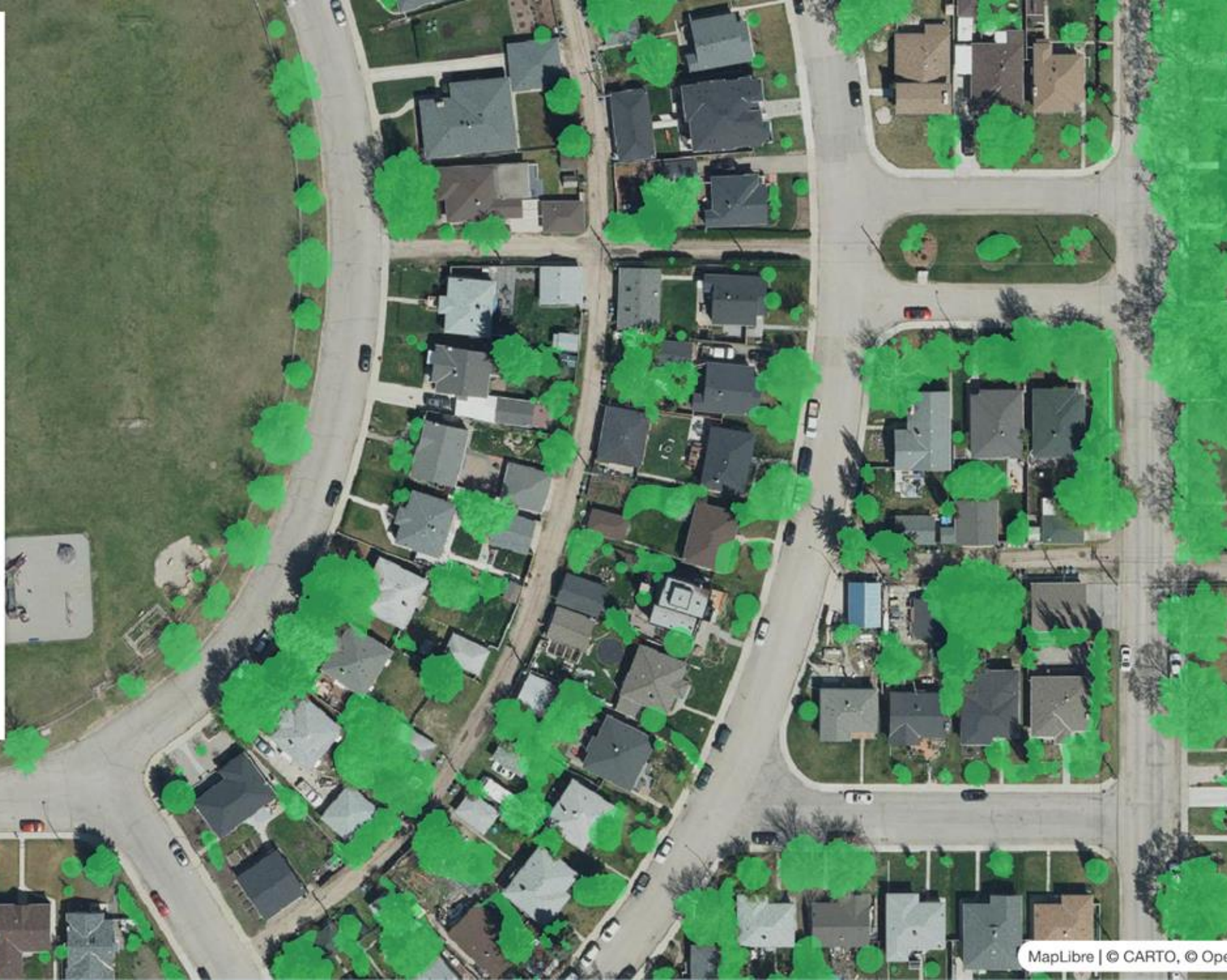
The percentage of a community that is covered by tree canopy.

Aerial View

High resolution bird's eye view using aerial photography.

☒ Show individual tree outlines

 Find out how the Tree Equity Score is calculated and data sources used for this tool.



Calgary Tree Equity



Tree Equity Canopy Cover Compare Years

Initial Year (Left)

2013

Comparison Year (Right)

2022

Canopy Cover Change (%)

The percentage of a community that is covered by tree canopy.

Aerial View

High resolution bird's eye view using aerial photography.

☒ Show individual tree outlines

[?](#) Find out how the Tree Equity Score is calculated and data sources used for this tool.

2013

2022



Climate Hub Tree Equity Campaign

- Promote equitable tree planting across Calgary and ensure sufficient budget allocations to support these plantings
- Advocate for tree equity policies with Calgary City Council
- Connect with community groups to advance tree equity engagement
- Celebrate and strengthen community stewardship with existing mini-forests



Thank You!

Github Repo: <https://github.com/smohiudd/calgary-tree-equity>